Evidence Analysis and Chain-of-Custody Report

Introduction

This practical exercise focused on analyzing digital evidence collected from a compromised system and maintaining proper chain-of-custody procedures. The objective was to identify suspicious network activity using Velociraptor and ensure forensic integrity using proper documentation techniques.

The investigation simulated a real-world scenario where a Windows system was suspected of unauthorized external communication.

Tools Used

Tool	Purpose
Velociraptor	Endpoint investigation and live forensic analysis
FTK Imager	Evidence acquisition and integrity verification

Evidence Collection

Evidence was collected from a Windows virtual machine suspected of malicious activity. Using Velociraptor, system artifacts and live network connection data were extracted.

FTK Imager was used to ensure proper acquisition and preservation of forensic evidence without altering original data.

Evidence Analysis (Velociraptor)

Query Executed

```
SELECT * FROM netstat
```

This query was used to retrieve all active and established network connections from the system.

Findings

The analysis revealed several active connections. Among them, the following were identified.

Observation

- Persistent connections to unknown external IP addresses
- Use of common ports (80, 443) to mask malicious traffic
- Repeated communication indicating possible command-and-control (C2) activity

Threat Analysis

The identified suspicious connections suggest:

- Possible malware beaconing to external servers
- Command-and-control (C2) communication
- Unauthorized data transmission

The external IPs should be further analyzed using threat intelligence platforms to confirm malicious intent.

Chain-of-Custody Documentation

Maintaining the integrity of evidence is critical in digital forensics. The following chain-of-custody record was created:

Item	Description	Collected By	Date	Hash Value
Network Log	Server-Z Log	SOC Analyst	2025-08-18	Given in screenshot

Key Points

- Evidence was collected in a controlled environment
- Hash value ensures data integrity
- Proper documentation ensures admissibility in investigations

Forensic Integrity Measures

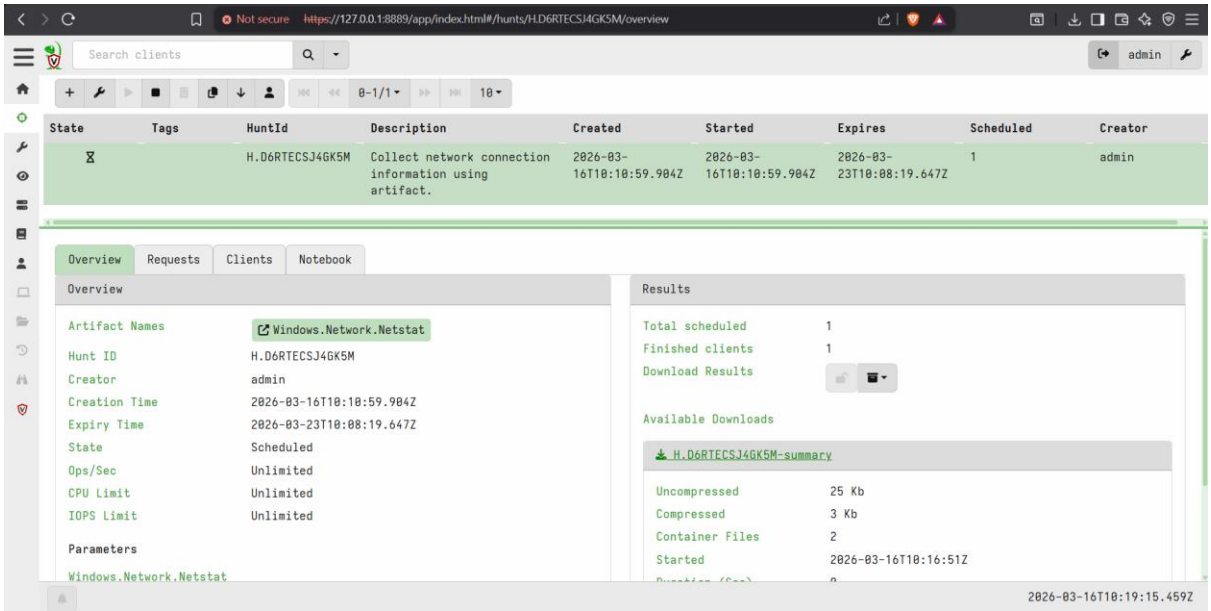
- No modification of original evidence
- Use of verified forensic tools (FTK Imager)

- Hash verification before and after analysis
- Secure storage of collected data

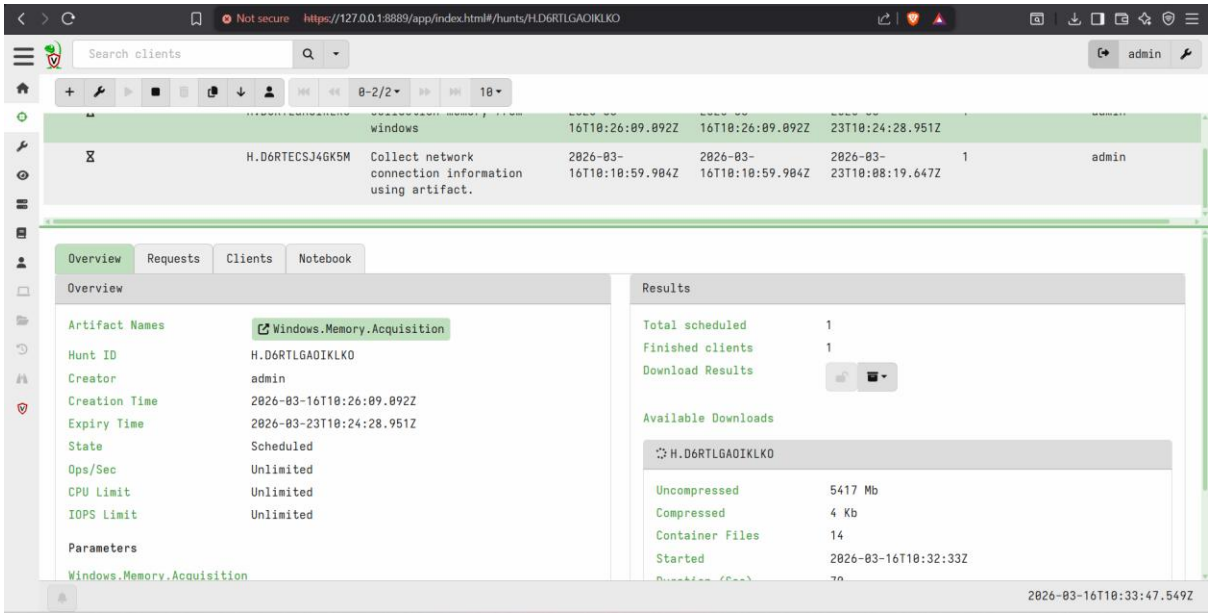
Conclusion

This practical exercise demonstrated the importance of evidence analysis and proper chain-of-custody in digital investigations. Using Velociraptor, suspicious network connections were identified, indicating potential malicious activity.

Maintaining forensic integrity through proper documentation and hashing ensures that evidence remains reliable and admissible. This highlights the critical role of structured forensic processes in cybersecurity investigations.



of Custody Report



```
C:\Users\USER\Downloads\memory_dump\results>certutil -hashfile "All Windows.Memory.Acquisition.csv" SHA256
SHA256 hash of All Windows.Memory.Acquisition.csv:
8d0ab37cb71ee6121385ca54a9d2e3d53b23afa4b95816a8da154d481d79c92f
CertUtil: -hashfile command completed successfully.

C:\Users\USER\Downloads\memory_dump\results>
```